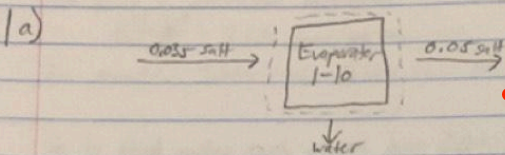


HW #4

- 1) Seawater has 3.5 wt% salt and mass flowrate $\dot{m}_F$. Fed into 10 evaporators. Brine leaving tenth evaporator contains 5 wt% salt and $\dot{m}_{10}$.



b) $\dot{m}_1 = \dot{m}_F - \frac{\dot{m}_W}{10}$

$\dot{m}_2 = \dot{m}_1 - \frac{\dot{m}_W}{10}$

$\dot{m}_3 = \dot{m}_2 - \frac{\dot{m}_W}{10}$

$\dot{m}_4 = \dot{m}_3 - \frac{\dot{m}_W}{10}$

$0.035 \dot{m}_F = 0.05 \dot{m}_{10}$

$\dot{m}_F = \dot{m}_W + \dot{m}_{10}$

Salt leaving solution in fourth evaporator: $w_{4,s} = \frac{0.035 \dot{m}_F}{\dot{m}_4}$

Fractional Yield fresh water from process: $F_y = \frac{\text{kg H}_2\text{O recovered}}{\text{kg H}_2\text{O original}}$

$F_y = \frac{\dot{m}_W}{(1 - 0.035) \dot{m}_F}$

c) $0.035 \dot{m}_F = 0.05 \dot{m}_{10}$, so $\dot{m}_{10} = \frac{0.035}{0.05} \dot{m}_F$, so $\dot{m}_{10} = 0.7 \dot{m}_F$

$\dot{m}_F = \dot{m}_W + \dot{m}_{10}$, so $\dot{m}_F = \dot{m}_W + 0.7 \dot{m}_F$, so $\dot{m}_W = 0.3 \dot{m}_F$

$\dot{m}_1 = \dot{m}_F - \frac{\dot{m}_W}{10}$, so $\dot{m}_1 = \dot{m}_F - \frac{0.3 \dot{m}_F}{10}$, so $\dot{m}_1 = \dot{m}_F - 0.03 \dot{m}_F$, so $\dot{m}_1 = 0.97 \dot{m}_F$

$\dot{m}_2 = \dot{m}_1 - \frac{\dot{m}_W}{10}$, so $\dot{m}_2 = 0.97 \dot{m}_F - 0.03 \dot{m}_F = 0.94 \dot{m}_F$

$\dot{m}_3 = \dot{m}_2 - \frac{\dot{m}_W}{10}$, so $\dot{m}_3 = 0.94 \dot{m}_F - 0.03 \dot{m}_F = 0.91 \dot{m}_F$

$\dot{m}_4 = \dot{m}_3 - \frac{\dot{m}_W}{10}$, so $\dot{m}_4 = 0.91 \dot{m}_F - 0.03 \dot{m}_F = 0.88 \dot{m}_F$

$w_{4,s} = \frac{0.035 \dot{m}_F}{\dot{m}_4}$, so $w_{4,s} = \frac{0.035 \dot{m}_F}{0.88 \dot{m}_F}$, so $w_{4,s} = 0.0398 = 3.98 \text{ wt\%}$

3.98 wt% salt mass leaving 4th evaporator

$F_y = \frac{\dot{m}_W}{(1 - 0.035) \dot{m}_F}$, so $F_y = \frac{0.3 \dot{m}_F}{0.965 \dot{m}_F}$, so $F_y = \frac{0.3}{0.965} = 0.3109 = 31.09\%$

31.09% is fractional yield of water

This problem doesn't need a basis for what's been asked so far. It would only need one to get actual flow rates

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1d) $\dot{m}_s$ is salt flow rate, w_F is salt mass fraction, $\dot{m}_F$ is mass flow

$\dot{m}_s = w_F \dot{m}_F$, so for any n evaporator: $\dot{m}_s = w_n \dot{m}_n$

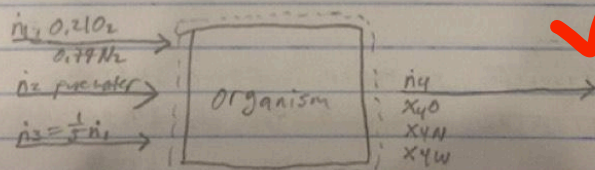
So brine flow after n^{th} evaporator: $\dot{m}_n = \dot{m}_F - \left(\frac{n \dot{m}_w}{10} \right)$

So mass salt fraction: $w_n = \frac{\dot{m}_s}{\dot{m}_n}$, so $w_n = \frac{w_F \dot{m}_F}{\dot{m}_F - \frac{n \dot{m}_w}{10}}$

Simplify: $w_n = \frac{w_F}{1 - \frac{n \dot{m}_w}{10 \dot{m}_F}}$, so $w_n = \frac{0.035}{1 - 0.03n}$

2) Humid air goes into organisms. Air (21 mol% O_2 , then balance N_2), liquid water at rate $20.0 \text{ cm}^3/\text{min}$, pure oxygen molar flow rate $\frac{1}{5}$ of molar flow rate of air input. Output is one single gas stream, with 1.5 mol% of water

2a)



2b) DoF Analysis

4a is 7 - n_1 has two comps, n_2 has one, n_3 has one, n_4 has three

4b is 0 - no independent chemical reactions

5a is 0 - no specified product flow

5b is 3 - n_1 is $2-1=1$ and n_4 is $3-1=2$

5c is 1 - $n_3 = \frac{1}{5} n_1$ is system specification

5d is 3 - water, O_2 , and N_2

$7 - (3+1+3) = 7-7=0$

Problem is specified

2c) Converting liquid water feed to moles/min: Use density then molar mass

• with Density: $\frac{20.0 \text{ cm}^3}{\text{min}} \cdot \frac{1 \text{ g}}{\text{cm}^3} = \frac{20 \text{ g}}{\text{min}}$

• with molar mass: $\frac{20 \text{ g}}{\text{min}} \cdot \frac{1 \text{ mol}}{18.015 \text{ g}} = 1.1102 \text{ mol/min}$ is $\dot{n}_2$

Water in gas leaving: $x_{w4} = 0.015$

• $\dot{n}_1 + \dot{n}_2 + \dot{n}_3 = \dot{n}_4$, so $\dot{n}_1 + \dot{n}_2 + \frac{1}{5}\dot{n}_1 = \dot{n}_4$, so $\frac{6}{5}\dot{n}_1 + \dot{n}_2 = \dot{n}_4$

• Balancing water: $x_{w4} = \frac{\dot{n}_2}{\dot{n}_4}$, so $0.015 = \frac{\dot{n}_2}{\frac{6}{5}\dot{n}_1 + \dot{n}_2}$

Solving for $\dot{n}_1$: $0.015(\frac{6}{5}\dot{n}_1 + \dot{n}_2) = \dot{n}_2$, so $\frac{6}{5}\dot{n}_1 + \dot{n}_2 = \frac{\dot{n}_2}{0.015}$
 $\frac{6}{5}\dot{n}_1 = \frac{\dot{n}_2}{0.015} - \dot{n}_2$, so $\dot{n}_1 = \frac{\dot{n}_2(\frac{1}{0.015} - 1)}{6/5}$, so $\dot{n}_1 = \frac{1.1102(66.67 - 1)}{1.2}$

$\dot{n}_1 = 60.76 \text{ mol/min}$

• Finding $\dot{n}_3$: $\dot{n}_3 = \frac{1}{5}\dot{n}_1$, so $\dot{n}_3 = \frac{1}{5}(60.76)$, so $\dot{n}_3 = 12.15 \text{ mol/min}$

• Finding $\dot{n}_4$: $\dot{n}_4 = \frac{6}{5}\dot{n}_1 + \dot{n}_2$, so $\dot{n}_4 = \frac{6}{5}(60.76) + 1.1102$, so $\dot{n}_4 = 74.02 \text{ mol/min}$

Element Flows

$\text{O}_2 = 0.21\dot{n}_1 + \dot{n}_3$, so $= 0.21(60.76) + 12.15 = 24.91 \text{ mol/min O}_2$

$\text{N}_2 = 0.79\dot{n}_1$, so $= 0.79(60.76) = 48.0 \text{ mol/min N}_2$

$\text{H}_2\text{O} = \dot{n}_2 = 1.1102 \text{ mol/min H}_2\text{O}$

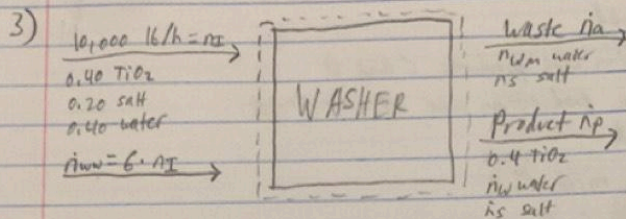
Mole fractions

$x_{\text{O}_2} = \frac{24.91}{24.91 + 48 + 1.1102}$, so $x_{\text{O}_2} = 0.3365$, or 33.65%

$x_{\text{N}_2} = \frac{48}{24.91 + 48 + 1.1102}$, so $x_{\text{N}_2} = 0.6485$, or 64.85%

$x_{\text{H}_2\text{O}} = 0.015$, or 1.50%

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- Washwater is 6 lb per pound of raw pigment, so $6 \cdot 10,000 \text{ lb/h} = 60,000 \text{ lb/h}$
- If 100% of TiO_2 in raw pigment is recovered in washed pigment, the 4,000 lb/h is recovered in washed pigment
- If washed pigment product leaving washer contains 50 wt% TiO_2 , total product = $\frac{\text{TiO}_2 \text{ in product}}{0.5}$
So $\frac{4,000}{0.5} = 8,000 \text{ lb/h}$
- Salt in product is $\frac{100 \text{ salt}}{1 \text{ million TiO}_2}$, so $4,000 \text{ lb TiO}_2 \cdot \frac{100}{1 \text{ mil}} = 0.4 \text{ lb/h salt}$
- Water in product: $8,000 - 4,000 - 0.4 = 3,999.6 \text{ lb/h}$
- Salt in waste: $2,000 - 0.4 = 1,999.6 \text{ lb/h}$
- Water in waste: $(4,000 + 60,000) - 3,999.6 = 60,000.4 \text{ lb/h}$
- Total waste: $1,999.6 + 60,000.4 = 62,000 \text{ lb/h}$
- Waste salt wt% = $\frac{1,999.6}{62,000.4} \cdot 100 = 3.225 \text{ wt\%}$

62,200 lb/h is too big for 30,000 lb/h limit
3.225 wt% salt is too big for 0.5 wt% limit

DOF analysis

- 4a is 9 - 3 in Ti , 1 in raw , 2 in 11a , 3 in 1p
- 4b is 0 - no independent chemical reactions
- 5a is 1 - 10,000 lb/h is specified product flow
- 5b is 4 - (3-1) in Ti , (3-1) in 1p , and (2-1) in 11a
- 5c is 1 - $\text{raw} = 6 \text{ in}$ is system specification
- 5d is 3 - TiO_2 , salt, and water

$9 - (1 + 4 + 1 + 3) = 0$, problem is well-specified

4) 4a is 24 - Counted

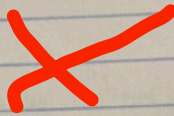
4b is 1 - Reaction with products C and D

5a is 1 - Given production rate of 25 g/mol/s

5b is 14 - Counted

5c is 1 - Molar ratio P, B

5d is 5 - P, I, B, C, D



$$(24+1) - (1+14+1+5) = 25 - 21 = 4, \text{ so under-specified}$$

DOF wrong, -5

No eqns setup, -15 pts

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